

Anina & shokoofeh

AI has not been used to solve the sheet.

1) Auto-correlation histogram (1 hour)

$$t_{\text{ref}} = 3\text{ms} \quad t \in [-20, 20\text{ms}] \quad \Delta = 0.1\text{ms} \quad T = 1\text{s} \quad \langle r \rangle = 100 \frac{\text{spikes}}{\text{s}}$$

$$\text{Driving rate Poisson process: } \langle r \rangle = \frac{\lambda}{1 + \lambda t_{\text{ref}}} \Leftrightarrow \lambda = \frac{\langle r \rangle}{1 - \langle r \rangle t_{\text{ref}}}$$

$$\lambda_{100} = \frac{100\text{Hz}}{1 - 100\text{Hz} \cdot 0.003\text{s}} = \frac{100\text{Hz}}{1 - 0.3} = \frac{100\text{Hz}}{0.7} = \frac{100 \frac{\text{spike}}{\text{s}}}{0.7} \approx 142.85\text{Hz}$$

$$\lambda_{300} = \frac{300\text{Hz}}{1 - 300\text{Hz} \cdot 0.003\text{s}} = \frac{300\text{Hz}}{1 - 0.9} = \frac{300\text{Hz}}{0.1} = 3000\text{Hz}$$

$$\lambda_{10} = \frac{10\text{Hz}}{1 - 10\text{Hz} \cdot 0.003\text{s}} = \frac{10\text{Hz}}{1 - 0.03} = \frac{10\text{Hz}}{0.97} \approx 10.3\text{Hz}$$

i) $N_m = 0$ during refractory period.So $0 < m < 3$ $N_m = 0$ | No spike pairs exist for lags smaller 3msii) Maximum value of N_m occurs directly after refractory

$$m = \pm \frac{t_{\text{ref}}}{\Delta t} = \frac{3\text{ms}}{0.1\text{ms}} = 30$$

Strength of peak increase with firing rate increase. (compensatory bump)

iii) Peaks: $N_{30} = r \cdot \Delta t$ $N_m = \langle r \rangle \cdot \Delta t$

$$\text{for } 10\text{Hz: } N_3 = 10 \cdot 10.3\text{Hz} \cdot 0.0001\text{s} = 0.00103 \text{ spike}$$

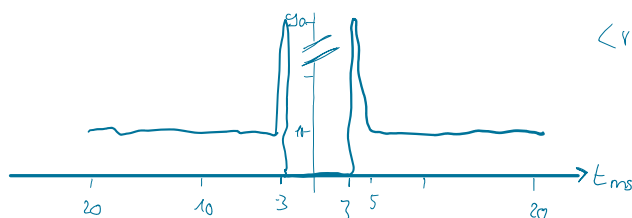
$$n \approx 10 \quad N_m = 10 \cdot 10\text{Hz} \cdot 0.0001\text{s} = 0.01 \text{ spike}$$

$$\text{for } 100\text{Hz: } N_3 = 100 \cdot 142.85\text{Hz} \cdot 0.0001\text{s} = 1.43 \text{ spike}$$

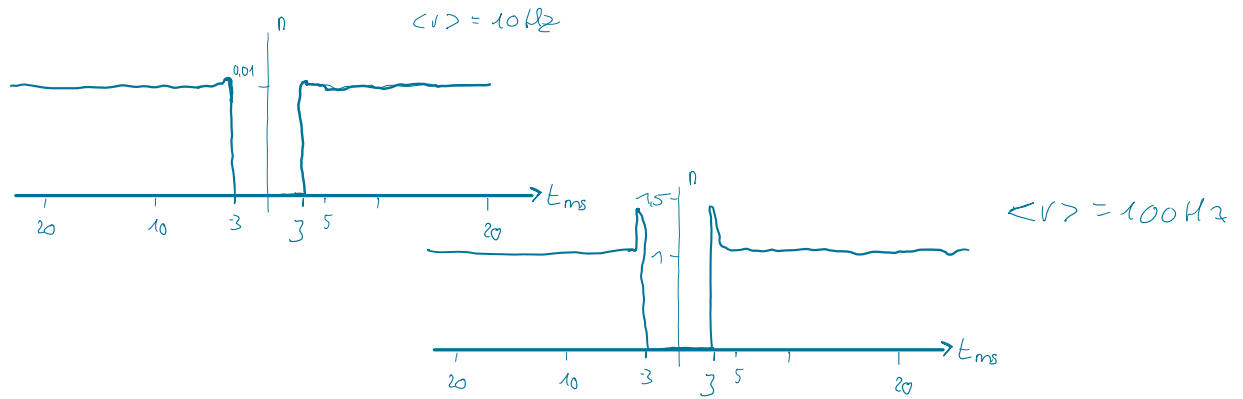
$$n = 100 \quad N_m = 100 \cdot 100\text{Hz} \cdot 0.0001\text{s} = 1 \text{ spike}$$

$$\text{for } 300\text{Hz: } N_3 = 300 \cdot 3000\text{Hz} \cdot 0.0001\text{s} = 90 \text{ spike}$$

$$n = 300 \quad N_m = 300 \cdot 300\text{Hz} \cdot 0.0001\text{s} = 9 \text{ spike}$$



$$\langle r \rangle = 300\text{Hz}$$



3. Symmetry of auto-correlation (20 min)

$$Q_{pp}(\tau) = Q_{pp}(-\tau)$$

$$Q_{pp}(\tau) := \frac{1}{T} \int_0^T dt \langle (p(t) - \langle r \rangle) \cdot (p(t+\tau) - \langle r \rangle) \rangle$$

$$t' = t + \tau \Rightarrow t = t' - \tau$$

$$dt' = dt$$

$$\text{Limits: } \begin{aligned} t=0 &\rightarrow t' = 0 + \tau = \tau \\ t=T &\rightarrow t' = T + \tau \end{aligned}$$

$$\frac{1}{T} \int_{\tau}^{T+\tau} \langle (p(t' - \tau) - \langle r \rangle) \cdot (p(t') - \langle r \rangle) \rangle dt'$$

For a homogenous poisson the expectation value only depend on the lag τ not the absolute time, so that the boundary corrections vanish.

$$\frac{1}{T} \int_0^T \langle (p(t' - \tau) - \langle r \rangle) \cdot (p(t') - \langle r \rangle) \rangle dt' = Q_{pp}(-\tau)$$

Moreover both spike times, t and $t \pm \tau$ must be inside the recording interval $[0, T]$, so actually t itself can just run up to $T - \tau$. As $\tau \ll T$ we can ignore this and set integral limits $[0, T]$ for t (and t')

2/ Cross-correlation of Oscillatory Signals (4 points) Time: 100 min

$$r_1(t) = [1 + \alpha_1 \sin(\omega t)] \bar{r}_1 \quad r_1, r_2: \text{constant firing rate}$$

$$r_2(t) = [1 + \alpha_2 \sin(\omega t + \phi)] \bar{r}_2 \quad \omega: \text{angular frequency}$$

ϕ : phase shift

$0 < \alpha_1, \alpha_2 < 1$: dimensionless constant

Calculate cross-correlation between r_1, r_2 .

Approximate ϕ for the case in which the integration time T is much longer than the length of one oscillation cycle $T \gg 2\pi/\omega$.

The cross-correlation for firing rates is:

$$Q_{r_1 r_2}(\tau) = \frac{1}{T} \int_0^T dt \, r_1(t) r_2(t + \tau) - \bar{r}_1 \bar{r}_2$$

Shift the second signal by τ .

$$r_2(t + \tau) = [1 + \alpha_2 \sin(\omega(t + \tau) + \phi)] \bar{r}_2$$

Now substitute both signals into the definition:

$$Q_{r_1 r_2}(\tau) = \frac{1}{T} \int_0^T dt \, [1 + \alpha_1 \sin(\omega t)] \bar{r}_1 [1 + \alpha_2 \sin(\omega(t + \tau) + \phi)] \bar{r}_2 - \bar{r}_1 \bar{r}_2$$

Factor out the constant:

$$Q_{r_1 r_2}(\tau) = \frac{1}{T} \bar{r}_1 \bar{r}_2 \int_0^T [1 + \alpha_1 \sin(\omega t)] [1 + \alpha_2 \sin(\omega t + \omega \tau + \phi)] dt - \bar{r}_1 \bar{r}_2$$

Now expand the product: $[1 + \alpha_1 \sin(\omega t)] [1 + \alpha_2 \sin(\omega t + \omega \tau + \phi)]$

$$1 + \alpha_2 \sin(\omega t + \omega \tau + \phi) + \alpha_1 \sin(\omega t) + \alpha_1 \alpha_2 \sin(\omega t) \sin(\omega t + \omega \tau + \phi)$$

Therefore:

$$\phi_{r_1 r_2}^* (\tau) = \bar{r}_1 \bar{r}_2 \left[\underbrace{\frac{1}{T} \int_0^T 1 dt}_{1} + \underbrace{\frac{a_2}{T} \int_0^T \sin(\omega t + \omega \tau + \phi) dt}_0 + \underbrace{\frac{a_1}{T} \int_0^T \sin(\omega t) dt}_0 + \underbrace{\frac{a_1 a_2}{T} \int_0^T \sin(\omega t) \sin(\omega t + \omega \tau + \phi) dt}_{\star} \right] - \bar{r}_1 \bar{r}_2$$

We assume that $T \gg 2\pi/\omega$, so the average of a sin over many cycle is approximately zero.

$$\frac{1}{T} \int_0^T \sin(\omega t) dt \approx 0, \quad \frac{1}{T} \int_0^T \sin(\omega t + \omega \tau + \phi) dt \approx 0$$

We have: $\sin(A) \sin(B) = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$

here we have: $A = \omega t$ $B = \omega t + \omega \tau + \phi$

So: $A - B = \cancel{\omega t} - \cancel{\omega t} - \omega \tau - \phi = -\omega \tau - \phi$

$$A + B = \omega t + \omega t + \omega \tau + \phi = 2\omega t + \omega \tau + \phi$$

Therefore $\cos(A-B) = \cos(-\omega \tau - \phi) = \cos(\omega \tau + \phi)$
[Cosine is even function]

Then we have:

$$\sin(\omega t) \sin(\omega t + \omega \tau + \phi) = \frac{1}{2} [\cos(\omega \tau + \phi) - \cos(2\omega t + \omega \tau + \phi)]$$

substitute to $\star$:

$$\begin{aligned} \int_0^T \sin(\omega t) \sin(\omega t + \omega \tau + \phi) dt &= \frac{1}{T} \int_0^T \frac{1}{2} [\cos(\omega \tau + \phi) - \cos(2\omega t + \omega \tau + \phi)] dt \\ &= \frac{1}{2T} \int_0^T \cos(\omega \tau + \phi) dt - \frac{1}{2T} \int_0^T \cos(2\omega t + \omega \tau + \phi) dt \end{aligned}$$

$$\underbrace{\frac{1}{2T} \overset{\text{constant}}{\cos(\omega\tau + Q)} \int_0^T dt}_{\text{first term}} - \underbrace{\frac{1}{2T} \times \frac{1}{2\omega} \sin(2\omega t + \omega\tau + Q) \Big|_0^T}_{\text{second term}}$$

$$\text{first term: } \frac{1}{2T} \cos(\omega\tau + Q) \times T = \frac{1}{2} \cos(\omega\tau + Q)$$

$$\text{second term: } \frac{1}{2T} \times \frac{1}{2\omega} \sin(2\omega T + \omega\tau + Q) - \sin(\omega\tau + Q)$$

$$|\sin(2\omega T + \omega\tau + Q) - \sin(\omega\tau + Q)| \leq 2 \quad \left(\begin{array}{l} \text{remember:} \\ -1 \leq \sin x \leq 1 \end{array} \right)$$

$$\text{Therefore: } \left| \frac{1}{T} \int_0^T \cos(2\omega t + \omega\tau + Q) dt \right| \leq \frac{1}{\omega T}$$

We assume: $T \gg \frac{2\pi}{\omega} \rightarrow T$ is very large the $\frac{1}{\omega T}$ is

very small. So we can approximate:

$$\frac{1}{T} \int_0^T \cos(2\omega t + \omega\tau + Q) dt \approx 0$$

$$\text{So we get: } \frac{1}{T} \int_0^T \sin(\omega t) \sin(\omega t + \omega\tau + Q) dt \approx \frac{1}{2} \cos(\omega\tau + Q)$$

Substitute all into * equation we have:

$$\phi_{r_1 r_2}(\tau) = \bar{r}_1 \bar{r}_2 \left[1 + \frac{a_1 a_2}{2} \cos(\omega\tau + Q) \right] - \bar{r}_1 \bar{r}_2$$

$$\bar{r}_1 \bar{r}_2 - \bar{r}_1 \bar{r}_2 = 0$$

So the final result is: $\frac{a_1 a_2 \bar{r}_1 \bar{r}_2}{2} \cos(\omega\tau + Q)$
→ amplitude
phase shift Q

$$\text{the peak: } \omega\tau + Q = 0 \Rightarrow \tau = -\frac{Q}{\omega} \leftarrow \text{maximum}$$

the interpretation is when second neuron shifts its phase relative to the first neuron by ϕ , then the cross-correlation peak appears at a time lag

$\tau = -\frac{\phi}{\omega}$. This lag tells us how much one oscillatory signal has to be shifted to align with the other.